

John M. Popovich, Jr. *, PT, DPT, PhD, Jacek Cholewicki, PhD, N. Peter Reeves, PhD, Lisa A. DeStefano, DO, Jacob J. Rowan, DO, Timothy J. Francisco, DO, Lawrence L. Prokop, DO, Mathew A. Zarkin, DO, Angela S. Lee, MPH, Alla Sikorskii, PhD, Pramod K. Pathak, PhD, Jongeun Choi, PhD, Clark J. Radcliffe, PhD and Ahmed Ramadan, PhD

The effects of osteopathic manipulative treatment on pain and disability in patients with chronic low back pain: a single-blinded randomized controlled trial

<https://doi.org/10.1515/jom-2022-0124>

Received June 16, 2022; accepted October 30, 2023;

published online January 11, 2024

Abstract

Context: The evidence for the efficacy of osteopathic manipulative treatment (OMT) in the management of low back pain (LBP) is considered weak by systematic reviews, because it is generally based on low-quality studies.

***Corresponding author: John M. Popovich, Jr.,** PT, DPT, PhD, Center for Neuromusculoskeletal Clinical Research, College of Osteopathic Medicine, Michigan State University, East Lansing, MI, USA; and Department of Osteopathic Manipulative Medicine, College of Osteopathic Medicine, Michigan State University, East Fee Hall, 965 Wilson Road, Room A439, East Lansing, MI 48824, USA, E-mail: popovi16@msu.edu

Jacek Cholewicki, PhD, **Lisa A. DeStefano,** DO, **Jacob J. Rowan,** DO, **Timothy J. Francisco,** DO, **Mathew A. Zarkin,** DO and **Angela S. Lee,** MPH, Center for Neuromusculoskeletal Clinical Research, College of Osteopathic Medicine, Michigan State University, East Lansing, MI, USA; and Department of Osteopathic Manipulative Medicine, College of Osteopathic Medicine, Michigan State University, East Lansing, MI, USA

N. Peter Reeves, PhD, Sumaq Life, LLC, East Lansing, MI, USA

Lawrence L. Prokop, DO, Center for Neuromusculoskeletal Clinical Research, College of Osteopathic Medicine, Michigan State University, East Lansing, MI, USA; and Department of Physical Medicine & Rehabilitation, College of Osteopathic Medicine, Michigan State University, East Lansing, MI, USA

Alla Sikorskii, PhD, Department of Psychiatry Osteopathic Medicine, College of Osteopathic Medicine, Michigan State University, East Lansing, MI, USA

Pramod K. Pathak, PhD, Department of Statistics and Probability, College of Natural Science, Michigan State University, East Lansing, MI, USA

Jongeun Choi, PhD, School of Mechanical Engineering, Yonsei University, Seoul, South Korea

Clark J. Radcliffe, PhD, Department of Mechanical Engineering, College of Engineering, Michigan State University, East Lansing, MI, USA

Ahmed Ramadan, PhD, Department of Biomedical Engineering, College of Science and Engineering, University of Minnesota, Minneapolis, MN, USA

Consequently, there is a need for more randomized controlled trials (RCTs) with a low risk of bias.

Objectives: The objective of this study is to evaluate the efficacy of an OMT intervention for reducing pain and disability in patients with chronic LBP.

Methods: A single-blinded, crossover, RCT was conducted at a university-based health system. Participants were adults, 21–65 years old, with nonspecific LBP. Eligible participants (n=80) were randomized to two trial arms: an immediate OMT intervention group and a delayed OMT (waiting period) group. The intervention consisted of three to four OMT sessions over 4–6 weeks, after which the participants switched (crossed-over) groups. The primary clinical outcomes were average pain, current pain, Patient-Reported Outcomes Measurement Information System (PROMIS) 29 v1.0 pain interference and physical function, and modified Oswestry Disability Index (ODI). Secondary outcomes included the remaining PROMIS health domains and the Fear Avoidance Beliefs Questionnaire (FABQ). These measures were taken at baseline (T_0), after one OMT session (T_1), at the crossover point (T_2), and at the end of the trial (T_3). Due to the carryover effects of OMT intervention, only the outcomes obtained prior to T_2 were evaluated utilizing mixed-effects models and after adjusting for baseline values.

Results: Totals of 35 and 36 participants with chronic LBP were available for the analysis at T_1 in the immediate OMT and waiting period groups, respectively, whereas 31 and 33 participants were available for the analysis at T_2 in the immediate OMT and waiting period groups, respectively. After one session of OMT (T_1), the analysis showed a significant reduction in the secondary outcomes of sleep disturbance and anxiety compared to the waiting period group. Following the entire intervention period (T_2), the immediate OMT group demonstrated a significantly better average pain outcome. The effect size was a 0.8 standard deviation (SD), rendering

the reduction in pain clinically significant. Further, the improvement in anxiety remained statistically significant. No study-related serious adverse events (AEs) were reported.

Conclusions: OMT intervention is safe and effective in reducing pain along with improving sleep and anxiety profiles in patients with chronic LBP.

Keywords: disability; low back pain; osteopathic manipulative medicine

Low back pain (LBP) is one of the most prevalent and costly health problems in Western society and is now the leading cause of disability globally [1, 2]. Chronic LBP, defined as pain lasting for 3 months or longer, is particularly burdensome. It has been estimated that over half a billion people across the globe suffer from chronic LBP on an annual basis [2]. The most common form of LBP is considered nonspecific [3, 4], which excludes LBP attributed to a specific pathology [5].

Clinical guidelines endorse a conservative, noninvasive approach to manage chronic, non-specific LBP [4, 6, 7]. Although there is variation among numerous international clinical practice guidelines, commonly recommended treatments are nonsteroidal anti-inflammatory and antidepressant drugs (NSAIDs), exercise therapy, and psychosocial interventions [8]. Several guidelines mention spinal manipulative therapy, but only 33 % of them recommend it as a treatment option for chronic LBP [8]. For example, based on the extensive literature review, the clinical guidelines from the North American Spine Society stated that evidence is conflicting whether the outcomes of spinal manipulative therapy for patients with chronic LBP are clinically different from no treatment, medication, or other modalities [9, 10]. However, this and similar reviews evaluating different manipulative techniques, such as chiropractic thrust manipulation, joint mobilization performed by physical therapists, and osteopathic manipulative treatment (OMT) performed by physicians, are all lumped under the umbrella of spinal manipulative therapy [9].

Osteopathic medicine is a distinct branch of medical practice in the United States, and physicians receive training in OMT that incorporates the therapeutic application of manual techniques as part of a whole-person approach to patient care [11]. OMT is utilized frequently for LBP management by osteopathic physicians in the United States [12]. OMT employs numerous manual techniques such as high-velocity low-amplitude (HVLA) thrust, soft tissue, myofascial release, muscle energy, articular, and counterstrain techniques, depending on the diagnosed somatic dysfunction and physician's preference [13, 14]. Based on 15 studies, the American Osteopathic Association guidelines state that OMT effectively reduces pain and

improves functional status in patients with acute and chronic nonspecific LBP [15], which is further corroborated by a systematic review and meta-analysis [16]. Nevertheless, the quality of evidence remains low, and more studies are necessary to strengthen the evidence [16]. Therefore, the purpose of this study is to further investigate the findings of existing studies on the efficacy of OMT in reducing pain and disability for chronic LBP with a randomized controlled trial (RCT). We hypothesize that OMT intervention will improve both pain and disability as measured in patients with chronic, nonspecific LBP.

Methods

Trial design

The original purpose of this study was to validate a novel seated balance test to quantify postural control as an objective measure of lumbar spine function [17–20]. Here, we separately present the clinical trial data involving patient-reported outcomes following OMT intervention. This was a sister study to an RCT on chronic neck pain with similar methodology including trial design, outcomes, and analyses [21]. A crossover design was adopted to maximize the number of data points for the validation while allowing all participants to receive OMT (Figure 1).

Adults with LBP were randomized to two trial arms: an immediate OMT intervention group (active intervention) and a delayed OMT intervention group (inactive control) with a 1:1 allocation ratio. Participants in the immediate OMT group received three to four OMT sessions (approximately once a week, allowing a minimum of 3 days and a maximum of 14 days between the OMT sessions) over a period of 4–6 weeks. The delayed OMT group did not receive any treatment for the first 4–6 weeks (i.e., waiting period), thus will be referred to as the waiting period group throughout the remainder of the article. After 4–6 weeks, patients switched (crossed-over) group assignments creating two treatment allocation sequences (AB and BA; Figure 1). Questionnaires with patient-reported outcomes were administered to all participants at baseline (T_0), after one OMT session or an equivalent time for the waiting period group (T_1), at a crossover time point (T_2), and at the end of the trial (T_3). Participants remained enrolled between 9 and 16 weeks to complete the entire study.

Before enrollment, the study was registered with ClinicalTrials.gov (NCT# 02261233) and approved by the Biomedical and Health Institutional Review Board (#14-212). The study had a Data and Safety Monitoring Board (DSMB) that consisted of two individuals not associated with the study institution and one individual from a different college at the study institution. Both the data and safety monitoring plan and DSMB were approved by the National Center for Complementary and Integrative Health (NCCIH). Additionally, a third-party monitor (Westat Corp., Rockville, MD) conducted pre-enrollment and subsequent annual site visits.

Participants

The eligible participants were adults with nonspecific LBP who met the inclusion/exclusion criteria listed in Table 1. Similarly, most clinical

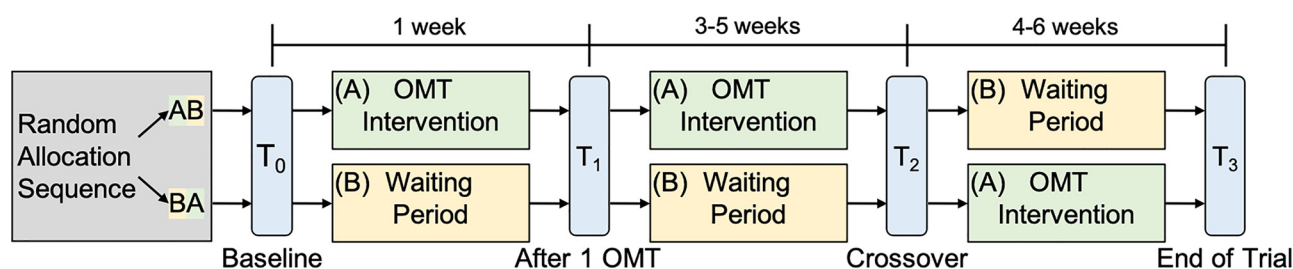


Figure 1: Schematic of the randomized, crossover, controlled trial design. Participants in the AB arm receive treatment (A – OMT intervention) followed by no treatment (B – waiting period), whereas participants in the BA arm receive no treatment (B – waiting period) followed by treatment (A – OMT intervention). Patient-reported outcomes were collected at baseline (T₀), after one OMT session (T₁), at the crossover point (T₂), and at the end of the trial (T₃).

guidelines define the “nonspecific” term by the exclusion of any LBP that is associated with serious local pathology or systemic disease (see exclusion criteria in Table 1) [7]. The eligibility of the participants was verified with a two-step process: (1) an online screening questionnaire that addressed most of the inclusion/exclusion criteria (a paper version was provided to those who did not have internet access); and (2) upon passing the online screening, participants signed an informed consent form and were examined by a physical medicine and rehabilitation (PM&R) board-certified physician (LLP) who ruled out “red flags” (e.g., history of trauma, history of cancer, weight loss, fever/chills/night sweats, infection, current status of being immunocompromised/prolonged corticosteroid use, bowel/bladder dysfunction, saddle anesthesia, night pain/intractable pain, infection, lower extremity neurological deficits) and verified the eligibility criteria. It should be noted that the original objective of validating a seated balance test as an objective treatment outcome did not necessitate symptom duration to be a part of the inclusion/exclusion criteria. However, only three participants did not meet the 3-month threshold for chronicity as defined in most studies [7]. To mitigate any potential concerns, in addition to the entire sample analysis, results for patients with chronic LBP only are also reported.

Study participants were recruited from the general population in the Greater Lansing (Michigan) and surrounding areas between May 2015 and June 2018. Recruitment strategies included personal communication, direct mail (utilizing numerous databases), emails and advertisements (e.g., newspaper, radio, handouts, flyers, websites, and social media). Flyers were also placed in clinics specializing in managing patients with musculoskeletal conditions. Participants were compensated with a \$100 gift card upon completion of the study.

Interventions

All OMT sessions were conducted at the Michigan State University (MSU) Osteopathic Manipulative Medicine Clinic or MSU Musculoskeletal Rehabilitation Clinic. Participants received OMT from one of five osteopathic physicians (LAD, JJR, TJF, MAZ, and LLP) specializing in OMT (three physicians board-certified in osteopathic neuromusculoskeletal medicine [ONMM] and two physicians board-certified in PM&R). These physicians initially evaluated participants for somatic dysfunction of the thoracic and lumbar spine, pelvis, and sacrum utilizing a protocol implemented by Licciardone et al [14]. Then, the physician treated the areas with somatic dysfunction utilizing OMT. The OMT techniques included a mandatory HVLA thrust technique to the lumbar spine region and any (or none) combination of the following four techniques: (i) soft tissue, (ii) muscle energy, (iii) myofascial, and (iv) articular. For patients who could not tolerate the HVLA treatment, a physician had to attempt this technique, minimally by attempting to place the patient

in the position to perform this maneuver. The physician re-evaluated the degree of somatic dysfunction during the treatment session and repeated or changed the technique. Treatment sessions were approximately 30 min in duration.

The OMT techniques utilized by treating physicians may vary for many reasons, including the degree of restriction, acuteness of symptoms, provider preference, or patient tolerance/cooperation [13]. However, no specific manual treatment technique demonstrates superior clinical outcomes, thus there is no optimal technique that can be recommended for treating LBP [22, 23]. Therefore, selecting an appropriate treatment protocol for our study was based on the most commonly utilized manual techniques by osteopathic physicians in the United States [13, 14, 24]. This protocol fits a more pragmatic research design, in which the efficacy of OMT was evaluated under the “usual treatment” conditions [25]. Nevertheless, implementing HVLA as a mandatory treatment and limiting physicians’ choices to four optional techniques made the treatment semi-standardized. For both groups, any non-study-related manipulation (chiropractic or osteopathic), physical therapy, massage, acupuncture, and spinal injections were prohibited throughout the duration of the study. However, participants were not required to adjust or modify existing medication regimens.

Outcomes

The primary clinical outcome measures were pain intensity and disability. Participants rated their current pain and the average pain over the last 7 days on an 11-point numeric rating scale (NRS) anchored with “no pain” at 0 and “worst pain imaginable” at 10. Disability was measured as a percentage utilizing the modified Oswestry Disability Index (ODI) [26] and Patient-Reported Outcomes Measurement Information System (PROMIS) 29 v1.0 domains of pain interference and physical function [27]. Secondary clinical outcome measures included the t scores for the remaining PROMIS health domains (satisfaction with participation in social roles, sleep disturbance, fatigue, depressive symptoms, and anxiety) and the Fear Avoidance Beliefs Questionnaire (FABQ) with both physical activity (FABQ-PA) and work (FABQ-W) subscales [28]. Patient-reported outcome measure questionnaires, along with additional items aimed at rechecking exclusion criteria, were administered online at each time point utilizing Research Electronic Data Capture (REDCap) [29] and in-person interviews. All source data collected in paper format were transcribed into the REDCap database utilizing a double data entry method.

Potential adverse events (AEs), including increased pain and discomfort that are common following OMT, were monitored

Table 1: Inclusion and exclusion criteria.**Inclusion Criteria**

- Age 21–65 years
- Independently ambulatory
- Able to speak and read english
- Able to understand study procedures and to comply with them for the entire length of the study
- Willing to be randomized to either immediate OMT or delayed OMT intervention groups
- Musculoskeletal pain: primarily in the low back region (from the lowest rib to the gluteal fold that may at times extend as referred pain into the thigh [10])
- Pain rating equal to or greater than 3/10 as indicated on the numeric rating scale for pain
- ODI equal to or greater than 26 %

Exclusion criteria

- Inability or unwillingness of the individual to give written informed consent
- Physical therapy or any other form of manual medicine (e.g., OMT, chiropractic manipulation, etc.), acupuncture or spinal injections within 1 month prior to study enrollment
- Workers' compensation benefits in the past 3 months or ongoing medical legal issues
- Possibly pregnant
- BMI>32
- Currently utilizing electrical implants (e.g., cardiac pacemakers, drug delivery pumps, etc.)
- History of:
 - Spinal surgery
 - Spinal fracture
 - Spinal infection (e.g., osteomyelitis)
 - Cancer
- Unresolved symptoms from:
 - Head trauma
 - Inner ear infection with associated balance and coordination problems
 - Orthostatic hypotension
 - Uncontrolled hypertension
 - Vestibular disorder (e.g., vertigo)
- Current diagnosis of:
 - Significant spinal deformity (e.g., scoliosis>20°)
 - Ankylosing spondylitis
 - Spondylolisthesis grades III or IV
 - Cauda equina syndrome
 - Rheumatoid arthritis
 - Osteoporosis
 - Angina or congestive heart failure symptoms
 - Active bleeding or infection in the back
 - Blindness
 - Seizures
 - Neurologic disease (e.g., Parkinson's disease, multiple sclerosis, cerebral palsy, Alzheimer's disease, amyotrophic lateral sclerosis, stroke or transient ischemic attack in the past year, cervical dystonia)
- Conditions recognized by a physician at any time during the study:
 - Significant or worsening signs of neurologic deficit
 - Symptoms are not consistent with mechanical findings
 - Other conditions impeding protocol implementation

BMI, body mass index; ODI, Oswestry disability index; OMT, osteopathic manipulative treatment.

immediately after treatment and followed-up weekly via email, telephone, and in-person contacts. Participants rated any signs or symptoms on an 11-point NRS. An increase in symptom severity by more than two points served as a threshold for classifying the event as an AE. The principal investigator (JC) graded the AEs according to their relatedness to this project, expectancy, and severity utilizing the Common Terminology Criteria for Adverse Events (CTCAE v4.03) [30].

Sample size

This study was powered for a medium effect size of 0.67 (Cohen's *d*) between patients with LBP and healthy individuals in trunk motor control tests. To detect these differences with a power of 0.80 or greater in two-sided tests at the 0.05 significance level, a sample of 36 participants in each group was required. This target was exceeded with 40 participants randomized to the immediate OMT group and 40 to the waiting period group (Figures 1 and 2).

Because carryover effects were present in the primary outcome measures (see Results), statistical analyses were performed utilizing data prior to the crossover allocation (T_1 and T_2) (Figure 1). Given the available sample size at T_2 of 31 participants in the immediate OMT group and 33 participants in the waiting period group, the effect size of Cohen's $d=0.71$ was detectable in unadjusted analyses as statistically significant with a power of 0.80 or greater in two-sided tests at 0.05 level of significance. In the analyses with adjustment for baseline, because of a correlation of approximately 0.35 between baseline and T_2 measures, the error variance was reduced, and the detectable effect size was $d=0.67$.

Randomization and blinding

A randomization module in REDCap was utilized to assign participants into the two groups. The allocation table was generated by a computer and locked once the project started. REDCap revealed the group assignment for one participant at the time of enrollment. Therefore, there was no way to predict any participant's allocation before enrollment or change group allocation afterward.

The study investigators, treating team physicians, and statisticians were all blinded to group assignment. Only the study coordinator and research assistants involved in coordinating clinical treatment had knowledge of group assignment. Study participants were instructed not to discuss their group assignment with the treating physicians and other study personnel.

Statistical methods

Participants' characteristics and outcome measures at baseline were summarized with descriptive statistics. Carryover effects were evaluated by comparing the summed outcomes for two time periods (T_0 to T_2 and T_2 to T_3) between the two allocation sequences (AB and BA) (Figure 1). Because tests of carryover effects are typically not powered [31], we estimated the effect sizes (Cohen's *d*) and applied a commonly utilized cutoff of $d=0.33$ for the clinical importance of the differences in patient-reported outcome data [32]. The characteristics of dropouts were compared by trial arm using *t* or chi-square tests as appropriate.

The unadjusted estimates of intervention effects on the outcomes were obtained from the *t*-tests that compared the means of the two trial arms at T_1 and T_2 . The adjusted estimates of the intervention effects were obtained from linear mixed-effects models relating repeated measures of the outcomes at T_1 and T_2 to the trial arm and baseline value of the

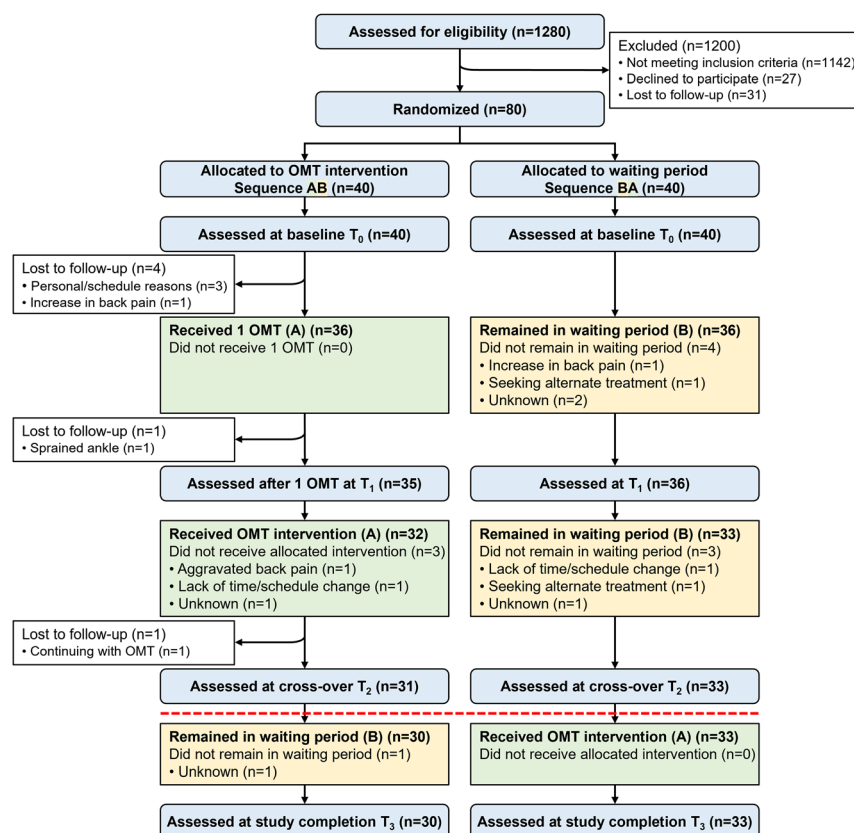


Figure 2: CONSORT flow diagram of participants. Because of the carryover effects in the primary outcomes, the comparison between study groups was carried out prior to crossover allocation at T_2 (indicated with a red dashed line), with 35 participants receiving immediate OMT intervention and 36 participants in the waiting period group at T_1 , and with 31 participants receiving immediate OMT intervention and 33 participants in the waiting period group at T_2 .

outcome, time (1 or 2), and time by trial arm interaction to evaluate potentially changing intervention effects as time progresses. The adjusted estimates utilize baseline values as a covariate, a recommended approach for RCTs [33–35], which aims to control for subject heterogeneity, thereby improving statistical efficiency for estimating treatment effects. The least square (LS) means according to the levels of the interaction were output from these models, and differences between them by trial arm were tested at each time point. The adjusted effect sizes were estimated as differences between LS means divided by the square root of the residual variance. Improvements in outcomes were clinically meaningful if the effect sizes for arm differences exceeded 0.5 of the standard deviation (SD) [32, 36]. The analyses were repeated removing the participants whose pain duration did not meet the definition of chronicity to evaluate the robustness of the findings.

Results

The recruitment and enrollment for this study ended in June 2018. The targeted accrual of 36 participants per trial arm was exceeded. Out of 1,280 screened volunteers, 80 met the eligibility criteria, agreed to participate, and were randomized (Figure 2). On average, participants in this study were 44

years old (range, 21–65 years), suffered from LBP for more than 12 years with an average pain rating of nearly six on a 0–10 scale, and the majority were women (Table 2).

Participants received a total of 261 OMT sessions. HVLA to the lumbar region was performed, or at least attempted, in 98.1 % of sessions ($n=256/261$), with cavitation occurring 55.1 % ($n=141/256$) of the time. There were two sessions in which somatic dysfunction was not identified in the lumbar region, thus HVLA was not attempted, and there were three sessions in which HVLA was not attempted as per protocol or not documented as such (i.e., protocol deviations occurred). In addition to HVLA, patients also received muscle energy (97.3 %), articulatory (21.1 %), myofascial (12.6 %), and soft tissue (4.6 %) treatment techniques across the lower thoracic, lumbar, sacrum, pelvis, and lower extremity regions.

There were carryover effects present in the primary outcomes (persistence of improvements in outcomes following the OMT intervention) for PROMIS pain interference ($p=0.16$, $d=0.35$) and PROMIS physical function ($p=0.06$, $d=0.48$), as well as in the secondary outcomes of FABQ-PA ($p=0.02$, $d=0.57$), PROMIS depression ($p=0.10$,

$d=0.42$), and PROMIS anxiety ($p=0.14$, $d=0.37$) profiles. Therefore, subsequent analyses were limited to the T_2 time point prior to the crossover allocation. During the time from T_0 to T_2 , nine participants dropped out from the immediate OMT group and seven from the waiting period group; however, their characteristics did not differ (Table 3). Consequently, 31 participants in the immediate OMT group and 33 participants in the waiting period group were available for comparison (Figure 2).

The between-group comparison following one OMT session (T_1) revealed immediate effects in the secondary outcomes favoring the immediate OMT group for PROMIS profiles of sleep disturbance and anxiety in the adjusted analyses (Table 4A). The immediate OMT group also had a better outcome for PROMIS physical function than the waiting period group in the unadjusted analysis. However, this result was most likely due to the trial arm differences at baseline (Table 2), as the between-arm difference was not significant in the adjusted analysis (Table 4A).

Following the entire OMT intervention period (T_2), in comparison to the waiting period group, the immediate OMT group showed a significantly better primary outcome for average pain, while also maintaining the improvements gained after one OMT session in the secondary outcomes of PROMIS sleep disturbance and anxiety (Table 4B). The unadjusted analyses returned significant group differences for PROMIS pain interference and FABQ-PA, which were not statistically significant after adjusting for baseline values (Table 4B).

After removing three participants (one man from the immediate OMT group and two women from the waiting period group) whose LBP did not meet the definition of chronicity (duration of pain lasting for a minimum of 3 months or longer), the results remained similar to the entire sample. The initial session of OMT produced significantly better secondary outcomes for PROMIS sleep disturbance and anxiety than the waiting period group in the adjusted analyses (Table 5A). The unadjusted analysis showed a better outcome for PROMIS physical function, but after adjusting for baseline values, it did not reach statistical significance (Table 5A). After the entire intervention period, the immediate OMT group had a better primary outcome for average pain, while maintaining the improvement gained after one OMT session in the secondary outcome of PROMIS anxiety (Table 5B). In contrast to the results from the entire sample, PROMIS sleep disturbance was no longer significantly better for the immediate OMT group when compared to the waiting period group, although the effect size of 0.5 SD is likely to be clinically significant (Table 5B). The effect sizes for the group differences among chronic LBP participants were 0.8 SD for the primary outcome of average pain and 1.1 SD for the

Table 2: Baseline characteristics of participants by trial arm.

Characteristic	Arm 1 (sequence AB: OMT then waiting period) n=40 Mean (SD) or n (%)	Arm 2 (sequence BA: waiting period then OMT) n=40 Mean (SD) or n (%)
Age	44.30 (13.03) [range: 21–65]	43.55 (11.97) [range: 24–64]
Sex		
Women	24 (60.0 %)	25 (62.5 %)
Men	16 (40.0 %)	15 (37.5 %)
Height, m	1.69 (0.08)	1.68 (0.08)
Weight, kg	76.18 (14.97)	78.55 (14.70)
BMI	26.51 (4.25)	27.72 (3.91)
Duration of LBP, years	12.48 (12.35)	12.96 (10.98)
Average pain	5.98 (1.75)	5.80 (1.83)
Current pain	5.13 (1.81)	5.50 (1.77)
ODI, %	35.3 (11.8)	30.7 (10.6)
FABQ work	17.33 (10.46)	11.80 (8.68)
FABQ physical activity	14.75 (4.86)	13.00 (5.25)
PROMIS profile pain interference	63.43 (5.13)	60.78 (4.99)
PROMIS profile satisfaction with participation in social roles	42.17 (6.34)	44.95 (6.68)
PROMIS profile sleep disturbance	56.66 (8.68)	53.72 (7.11)
PROMIS profile fatigue	56.07 (9.38)	54.56 (8.03)
PROMIS profile depression	49.70 (8.20)	49.65 (7.83)
PROMIS profile anxiety	54.34 (9.74)	50.10 (8.44)
PROMIS profile physical function	39.63 (4.35)	42.77 (6.37)

BMI, body mass index; FABQ, fear avoidance beliefs questionnaire; LBP, low back pain; ODI, Oswestry disability index; OMT, osteopathic manipulative treatment; PROMIS, patient-reported outcomes measurement information system; SD, standard deviation.

secondary outcome of PROMIS anxiety, which also falls in the clinically significant range [32, 36].

There were 122 study-related AEs reported during this study, and none of them were rated as serious (Severity Grade 4 or greater). One unrelated serious AE, colitis that required hospitalization, was also reported. Of the 122 study-related AEs, only 31 AEs could be attributed to any of the 261 delivered OMT sessions by classifying them as at least “possibly related” (Relatedness Grade 2 or greater). The other AEs were either not related to the study (Relatedness Grade 0 or 1) or were associated with the laboratory-based postural control testing sessions. All 31 AEs were mild or moderate (Grade 1 or 2) increases in LBP ($n=10$), muscle soreness/tightness ($n=13$), stiffness/achiness ($n=3$), and other ($n=5$). Almost all OMT-related AEs resolved completely, one with minor sequela.

Table 3: Demographic characteristics and primary outcome values of dropouts from baseline (T_0) to crossover point (T_2) by study group.

Characteristic	OMT group n=9 Mean (SD) or n (%)	Waiting period group n=7 Mean (SD) or n (%)	p-Value
Age	40.11 (13.04) [range: 21–64]	40.14 (6.87) [range: 30–52]	0.99
Sex			0.77
Women	7 (77.8 %)	5 (71.4 %)	
Men	2 (22.2 %)	2 (28.6 %)	
Duration of LBP, years	11.00 (11.93)	10.00 (7.63)	0.85
Average pain	6.00 (2.18)	6.00 (2.00)	0.99
Current pain	5.33 (1.94)	6.14 (1.68)	0.39
ODI, %	37.1 (16.7)	36.0 (11.3)	0.85
PROMIS profile pain interference	65.37 (5.30)	62.71 (3.35)	0.27
PROMIS profile physical function	39.26 (4.80)	40.67 (4.02)	0.54

LBP, low back pain; ODI, Oswestry disability index; OMT, osteopathic manipulative treatment; PROMIS, patient-reported outcomes measurement information system; SD, standard deviation.

Discussion

The results from the current study indicate that OMT intervention is effective in reducing pain in patients with chronic LBP, as compared to a waiting period control group. The significant trial arm differences in the primary outcome in this study was 1.3 points for average pain NRS scores in the adjusted means. It is debated how much improvement in pain can be considered clinically significant or important. Typically, the minimal clinically important difference (MCID) is calculated as the smallest difference that patients perceive to be beneficial [37, 38]. However, the reported MCID values are inconsistent across different studies and vary widely [39]. For that reason, Norman et al. [36] proposed, and others supported [32], the 0.5 SD as a conservative estimate of an effect size that is likely to be clinically meaningful. By this measure, the statistically significant improvement in average pain of 0.8 SD observed in this study appears to be clinically meaningful.

The participants receiving OMT (i.e., immediate OMT group) also showed significant improvements in secondary

Table 4A: Outcomes for all participants at T_1 time point, following one OMT session or an equivalent waiting period, by study group.

Outcome	Unadjusted			Adjusted		
	OMT group Mean (SE)	Waiting period group Mean (SE)	p-Value (95 % CI for the difference), effect size	OMT group LS mean (SE)	Waiting period group LS mean (SE)	p-Value (95 % CI for the difference), effect size
Average pain	5.29 (0.29)	5.39 (0.31)	0.80 (−0.95, 0.75), d=0.05	5.06 (0.29)	5.61 (0.28)	0.18 (−1.37, 0.27), d=0.33
Current pain	4.66 (0.35)	4.94 (0.37)	0.57 (−1.30, 0.73), d=0.13	4.56 (0.31)	5.03 (0.31)	0.30 (−1.38, 0.43), d=0.27
ODI, %	33.3 (1.9)	29.7 (2.0)	0.18 (−1.7, 9.0), d=0.32	30.4 (1.3)	32.1 (1.3)	0.38 (−5.6, 2.1), d=0.22
FABQ work	14.46 (1.62)	12.67 (1.44)	0.41 (−2.53, 6.11), d=0.19	13.04 (1.07)	13.84 (1.05)	0.61 (−3.86, 2.27), d=0.13
FABQ physical activity	15.06 (0.71)	13.33 (0.81)	0.11 (−0.42, 3.88), d=0.38	14.51 (0.62)	14.00 (0.60)	0.56 (−1.26, 2.28), d=0.14
PROMIS profile pain interference	60.98 (1.01)	59.53 (0.91)	0.28 (1.26, 4.16), d=0.25	59.70 (0.85)	60.59 (0.83)	0.46 (−3.30, 1.52), d=0.18
PROMIS profile satisfaction with participation in social roles	43.15 (1.08)	44.34 (1.39)	0.49 (−4.79, 2.32), d=0.16	44.98 (1.03)	42.93 (1.01)	0.17 (−0.88, 4.99), d=0.35
PROMIS profile sleep disturbance	54.29 (1.61)	54.88 (1.31)	0.77 (−4.71, 3.54), d=0.07	52.89 (0.95)	56.16 (0.93)	0.02 (−5.98, −0.55), d=0.60
PROMIS profile fatigue	55.97 (1.65)	55.49 (1.75)	0.84 (−4.32, 5.29), d=0.05	54.98 (1.32)	56.39 (1.30)	0.45 (−5.19, 2.36), d=0.18
PROMIS profile depression	48.32 (1.46)	49.97 (1.57)	0.45 (−5.93, 2.63), d=0.18	48.59 (0.91)	49.39 (0.90)	0.54 (−3.43, 1.82), d=0.15
PROMIS profile anxiety	50.17 (1.51)	50.35 (1.51)	0.93 (−4.45, 4.08), d=0.02	48.45 (1.01)	52.04 (0.04)	0.02 (−6.48, −0.70), d=0.62
PROMIS profile physical function	40.18 (0.65)	43.07 (1.15)	0.03 (−5.54, −0.24), d=0.36	41.36 (0.74)	41.97 (0.73)	0.57 (−2.71, 1.50), d=0.14

CI, confidence interval; FABQ, fear avoidance beliefs questionnaire; ODI, Oswestry disability index; OMT, osteopathic manipulative treatment; PROMIS, patient-reported outcomes measurement information system; SE, standard error. Statistically significant ($p < 0.05$) differences are bolded.

Table 4B: Postintervention outcomes for all participants at T₂ time point by study group.

Outcome	Unadjusted			Adjusted		
	OMT group Mean (SE)	Waiting period group Mean (SE)	p-value (95 % CI for the difference), effect size	OMT group LS mean (SE)	Waiting period group LS mean (SE)	p-value (95 % CI for the difference), effect size
Average pain	4.42 (0.40)	5.42 (0.38)	0.07 (−2.11, 0.10), d=0.45	4.20 (0.30)	5.63 (0.30)	0.0016 (−2.29, −0.56), d=0.86
Current pain	4.35 (0.42)	5.09 (0.40)	0.21 (−1.89, 0.41), d=0.32	4.28 (0.33)	5.20 (0.32)	0.06 (−1.87, 0.004), d=0.51
ODI, %	31.9 (2.2)	29.1 (2.3)	0.38 (−3.6, 9.2), d=0.23	29.3 (1.4)	31.9 (1.4)	0.21 (−6.6, 1.5), d=0.32
FABQ work	13.58 (1.97)	11.21 (1.53)	0.34 (−2.58, 7.32), d=0.24	12.21 (1.12)	12.60 (1.09)	0.81 (−3.58, 2.80), d=0.06
FABQ physical activity	15.52 (0.67)	12.45 (0.90)	0.01 (0.70, 5.42), d=0.65	14.72 (0.65)	13.09 (0.63)	0.08 (−0.21, 3.47), d=0.46
PROMIS profile pain interference	62.13 (1.24)	58.51 (1.04)	0.03 (0.39, 6.84), d=0.56	61.03 (0.89)	59.75 (0.87)	0.31 (−1.30, 3.81), d=0.26
PROMIS profile satisfaction with participation in social roles	42.09 (1.33)	43.01 (1.32)	0.62 (−4.68, 2.83), d=0.13	43.87 (1.09)	41.06 (1.06)	0.07 (−0.29, 5.90), d=0.47
PROMIS profile sleep disturbance	53.81 (1.59)	53.80 (1.38)	0.99 (−4.19, 4.21), d=0.00	52.30 (1.00)	55.40 (0.97)	0.03 (−5.94, −0.27), d=0.56
PROMIS profile fatigue	55.90 (1.92)	55.47 (1.69)	0.86 (−4.64, 5.51), d=0.04	54.93 (1.38)	56.48 (1.34)	0.43 (−5.47, 2.37), d=0.26
PROMIS profile depression	45.91 (1.27)	48.72 (1.58)	0.18 (−6.87, 1.28), d=0.34	46.37 (0.94)	48.64 (0.92)	0.10 (−4.96, 0.43), d=0.43
PROMIS profile anxiety	47.37 (1.44)	49.60 (1.56)	0.30 (−6.48, 2.01), d=0.27	45.22 (1.05)	51.45 (1.02)	0.0001 (−9.22, −3.23), d=1.07
PROMIS profile physical function	40.03 (0.92)	42.08 (1.02)	0.14 (−4.82, 0.71), d=0.37	41.30 (0.78)	40.88 (0.77)	0.71 (−1.79, 2.62), d=0.10

CI, confidence interval; FABQ, fear avoidance beliefs questionnaire; ODI, Oswestry disability index; OMT, osteopathic manipulative treatment; PROMIS, patient-reported outcomes measurement information system; SE, standard error. Statistically significant ($p < 0.05$) differences are bolded.

outcomes manifested in reduced anxiety scores. It is worth noting that improvement in anxiety occurred immediately after one OMT session, whereas reduction in average pain took longer (three to four sessions). This is consistent with the Burns et al. [40] study of 65 chronic pain patients, in which early-treatment reductions in catastrophizing and pain-related anxiety predicted late-treatment improvements in pain severity. An improvement in anxiety scores was not proposed as a primary outcome because the theorized mechanisms of manual treatment do not typically include modifications of psycho-social factors [41–43]. However, there is a well-established link between chronic pain and depressive disorders, which share similar neurophysiological pathways [44, 45]. Sleep disturbance, pain, anxiety, depression, and fatigue (lack of energy), known as the SPADE cluster, often co-occur in the general population and are difficult to manage in a primary clinical practice [46–48]. Concomitant improvements in pain, anxiety, and sleep disturbance in the group receiving OMT intervention

demonstrate the efficacy of OMT in addressing SPADE symptoms and point to the consistency in the results from the current study. Considering that the holistic, whole-person approach to treating patients is embraced by the osteopathic profession, such findings highlight the potential for OMT to affect multiple mechanistic pathways involved in treating neuromusculoskeletal conditions, particularly multifactorial [49] conditions like LBP.

There is likely a dose-response relationship to OMT intervention. Although there are no studies addressing specifically the dose response of OMT for chronic LBP, the responder analysis in a large RCT ($n=455$) suggests that six OMT sessions are necessary to obtain a substantial outcome in which 50 % of patients achieve at least 50 % pain improvement [50]. Therefore, it is possible that more than four OMT sessions would result in even greater pain reduction. Nevertheless, the findings observed in the primary and secondary outcomes of this study provide additional evidence that OMT is an effective treatment for patients with

Table 5A: Outcomes for chronic LBP participants only at T₁ time point, following 1 OMT session or an equivalent waiting period, by study group.

Outcome	Unadjusted			Adjusted		
	OMT group Mean (SE)	Waiting period group Mean (SE)	p-Value (95 % CI for the difference), effect size	OMT group LS mean (SE)	Waiting period group LS mean (SE)	p-Value (95 % CI for the difference), effect size
Average pain	5.24 (0.30)	5.32 (0.32)	0.84 (−0.96, 0.79), d=0.05	5.00 (0.30)	5.56 (0.30)	0.19 (−1.42, 0.29), d=0.33
Current pain	4.59 (0.35)	4.88 (0.39)	0.58 (−1.34, 0.75), d=0.14	4.45 (0.32)	5.00 (0.32)	0.24 (−1.49, 0.39), d=0.30
ODI, %	33.7 (1.9)	30.1 (2.0)	0.20 (−1.9, 9.1), d=0.31	30.8 (1.4)	32.6 (1.4)	0.38 (−5.7, 2.2), d=0.23
FABQ work	14.62 (1.66)	12.65 (1.51)	0.38 (−2.51, 6.45), d=0.21	13.08 (1.11)	13.97 (1.11)	0.58 (−4.11, 2.33), d=0.14
FABQ physical activity	15.09 (0.73)	13.24 (0.84)	0.10 (−0.37, 4.08), d=0.41	14.54 (0.64)	13.93 (0.63)	0.51 (−1.22, 2.46), d=0.17
PROMIS profile pain interference	61.10 (1.03)	59.43 (0.96)	0.24 (−1.15, 4.48), d=0.29	59.71 (0.88)	60.6 (0.88)	0.49 (−3.42, 1.65), d=0.18
PROMIS profile satisfaction with participation in social roles	42.90 (1.08)	44.00 (1.43)	0.54 (−4.68, 2.48), d=0.15	44.83 (1.05)	42.44 (1.05)	0.12 (−0.65, 5.43), d=0.40
PROMIS profile sleep disturbance	54.68 (1.60)	54.86 (1.38)	0.93 (−4.41, 4.04), d=0.02	53.08 (0.95)	56.39 (0.95)	0.02 (−6.06, −0.58), d=0.62
PROMIS profile fatigue	56.36 (1.66)	55.17 (1.72)	0.62 (−3.59, 5.96), d=0.12	55.30 (1.33)	56.17 (1.32)	0.65 (−4.70, 2.97), d=0.11
PROMIS profile depression	48.53 (1.48)	49.87 (1.6)	0.54 (−5.7, 3.02), d=0.15	48.66 (0.94)	49.41 (0.94)	0.59 (−3.49, 2.00), d=0.14
PROMIS profile anxiety	50.07 (1.55)	50.32 (1.54)	0.91 (−4.62, 4.11), d=0.03	48.51 (1.04)	51.88 (1.03)	0.03 (−6.37, −0.37), d=0.57
PROMIS profile physical function	40.03 (0.65)	43.05 (1.22)	0.03 (−5.79, −0.25), d=0.53	41.26 (0.76)	41.87 (0.76)	0.58 (−2.81, 1.58), d=0.14

CI, confidence interval; FABQ, fear avoidance beliefs questionnaire; ODI, Oswestry disability index; OMT, osteopathic manipulative treatment; PROMIS, patient-reported outcomes measurement information system; SE, standard error. Statistically significant ($p < 0.05$) differences are bolded.

LBP, and further endorse the American Osteopathic Association guidelines that recommend that osteopathic physicians utilize OMT as a nonpharmacological treatment for nonspecific acute and chronic LBP [15].

Although this study demonstrated an improvement in pain following OMT, there was no statistically significant improvement in the ODI or PROMIS physical functional domain scores after accounting for baseline differences (i.e., adjusted analysis). This finding is of interest because a primary goal of OMT is to restore function [51, 52], and systematic reviews concluded that OMT provides improvements in both pain and function in patients with chronic pain [16, 53]. As stated previously, we may have seen a greater effect in other patient-reported outcomes with more OMT sessions (i.e., dose response). It is also possible that no differences in ODI or PROMIS physical function scores were observed because this study was not powered to detect such differences. Our findings are more consistent with a large RCTs of 455 participants with LBP that also showed

improvements in pain, but not in Roland-Morris Disability Questionnaire scores when comparing patients receiving OMT vs. sham OMT [50]. Similarly, a pain registry study of 404 patients with chronic LBP showed that OMT led to reduced pain intensity but did not improve scores on the Roland-Morris Disability Questionnaire [54]. Therefore, future studies are warranted to determine how and for whom OMT intervention can be most beneficial.

In this study, we observed clinically meaningful reductions in primary and secondary patient outcomes of pain and anxiety, respectively; however, OMT has also been shown to have a significant effect on reducing medications in patients with LBP [50, 55–57]. For example, Licciardone and Gatchel [55] showed patients with chronic LBP ($n=79$) who were treated by osteopathic physicians who utilized OMT reported a significant decrease in the use of NSAIDs or opioid drugs. Further, a study by Montrose and colleagues [56] showed a reduction in cyclobenzaprine usage in 10 patients that were prescribed pain medications and received OMT. We collected

Table 5B: Postintervention outcomes for chronic LBP participants only, at T₂ time point, by study group.

Outcome	Unadjusted			Adjusted		
	OMT group Mean (SE)	Waiting period group Mean (SE)	p-Value (95 % CI for the difference), effect size	OMT group LS mean (SE)	Waiting period group LS mean (SE)	p-Value (95 % CI for the difference), effect size
Average pain	4.47 (0.41)	5.35 (0.40)	0.13 (−2.04, 0.26), d=0.40	4.24 (0.31)	5.58 (0.31)	0.0043 (−2.23, −0.43), d=0.79
Current pain	4.43 (0.42)	5.00 (0.42)	0.34 (−1.76, 0.62), d=0.24	4.33 (0.34)	5.15 (0.34)	0.10 (−1.8, 0.17), d=0.45
ODI, %	32.1 (2.2)	29.4 (2.4)	0.40 (−3.7, 9.3), d=0.22	29.6 (1.4)	32.2 (1.4)	0.22 (−6.7, 1.5), d=0.33
FABQ work	13.70 (2.03)	11.16 (1.61)	0.33 (−2.63, 7.71), d=0.25	12.19 (1.17)	12.73 (1.15)	0.75 (−3.89, 2.82), d=0.08
FABQ physical activity	15.50 (0.78)	12.29 (0.95)	0.01 (0.74, 5.68), d=0.67	14.69 (0.67)	12.96 (0.66)	0.07 (−0.18, 3.65), d=0.48
PROMIS profile pain interference	62.16 (1.28)	58.30 (1.10)	0.03 (0.49, 7.24), d=0.59	60.97 (0.93)	59.68 (0.92)	0.34 (−1.38, 3.95), d=0.26
PROMIS profile satisfaction with participation in social roles	41.77 (1.34)	42.9 (1.41)	0.56 (−5.01, 2.76), d=0.15	43.68 (1.11)	40.78 (1.1)	0.08 (−0.32, 6.12), d=0.48
PROMIS profile sleep disturbance	54.36 (1.55)	53.39 (1.42)	0.65 (−3.23, 5.16), d=0.12	52.59 (1.00)	55.28 (0.98)	0.07 (−5.56, 0.18), d=0.5
PROMIS profile fatigue	56.34 (1.94)	55.32 (1.76)	0.7 (−4.22, 6.26), d=0.1	55.29 (1.38)	56.42 (1.36)	0.57 (−5.11, 2.84), d=0.15
PROMIS profile depression	46.07 (1.30)	48.73 (1.65)	0.21 (−6.87, 1.56), d=0.32	46.38 (0.97)	48.82 (0.97)	0.09 (−5.26, 0.38), d=0.46
PROMIS profile anxiety	47.24 (1.48)	49.70 (1.62)	0.27 (−6.85, 1.93), d=0.29	45.27 (1.08)	51.44 (1.07)	0.0002 (−9.29, −3.05), d=1.05
PROMIS profile physical function	40.06 (0.95)	42.00 (1.09)	0.19 (−4.84, 0.96), d=0.34	41.39 (0.81)	40.71 (0.8)	0.56 (−1.63, 2.99), d=0.16

CI, confidence interval; FABQ, fear avoidance beliefs questionnaire; ODI, Oswestry disability index; OMT, osteopathic manipulative treatment; PROMIS, patient-reported outcomes measurement information system; SE, standard error. Statistically significant ($p < 0.05$) differences are bolded.

patient-reported medication usage as part of this project, but the data were inconsistently recounted by participants, and we did not have the ability to review the medical records of the general population sample utilized in this study. Consequently, these data were not reported here. Considering the clinically meaningful reductions in pain, we can only speculate that similar findings would have occurred in this study.

The lack of long-term follow-up is a major limitation of the current study, although the persistence of improvements in the primary clinical outcomes can be judged from the significant carryover effects that were sustained for at least 4 weeks in the second part of this crossover experimental design. Participants were recruited based on self-reported LBP and were not necessarily seeking care for their symptoms, which may not generalize to patients actively seeking care who might have shown larger effects. Furthermore, because this was a validation study for the assessment of postural control and lumbar spine function, it did not include an active treatment control group and was

based on a single health system. Nevertheless, because of the paucity of RCTs on the efficacy of OMT for patients with chronic LBP, this study provides additional data on this topic and will help in updating future clinical guidelines. Because osteopathic physicians in the current study were relatively free to choose techniques and areas of the body to treat based on their clinical examination, the results should have broader generalizability to other osteopathic medical practices.

Conclusions

In conclusion, our study demonstrated that OMT intervention effectively reduces pain and anxiety, along with improving sleep in patients with chronic LBP. Further, treatment with OMT in patients with LBP is safe because there were no study-related serious AEs. Therefore, OMT should be utilized to care for patients with LBP as recommended in the American Osteopathic Association guidelines.

Acknowledgments: The authors would like to thank Dr. M. Cody Priess for his significant role in the development of the seated balance testing apparatus. We would also like to extend our sincere gratitude and respects to our late colleague, Timothy J. Francisco, DO, whose dedication and significant contribution to the osteopathic profession were instrumental in the realization of this work.

Research ethics: This study was conducted in accordance with the Declaration of Helsinki (as revised in 2013). Prior to enrollment, the study was registered with ClinicalTrials.gov (NCT# 02261233) and approved by the Michigan State University Biomedical and Health Institutional Review Board (#14-212).

Informed consent: Informed consent was obtained from all individuals included in this study.

Author contributions: The authors have accepted responsibility for the entire content of this manuscript and approved its submission. All authors provided substantial contributions to conception and design, acquisition of data, or analysis and interpretation of data; all authors drafted the article or revised it critically for important intellectual content; all authors gave final approval of the version of the article to be published; and all authors agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Competing interests: The authors state no conflict of interest.

Research funding: This publication was made possible by grant number U19AT006057 from the National Center for Complementary and Integrative Health (NCCIH) at the National Institutes of Health. Its contents are solely the responsibility of the authors and do not necessarily represent the official views of NCCIH.

Data availability: De-identified data that support the findings of this study can be obtained from the corresponding author, upon reasonable request.

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